

BiteFixes Enterprise Architecture Manual

Volume 14

Data Architecture (DA)

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14.1 Purpose

The Data Architecture defines how information is created, organized, stored, protected, exchanged, and consumed throughout the BiteFixes Enterprise Platform.

Its objectives are to ensure:

- Consistency
- Integrity
- Scalability
- Availability
- Traceability
- Security
- Business intelligence readiness
- AI readiness

The architecture supports both current operational needs and future enterprise-scale growth.

14.2 Data Architecture Vision

Data is considered the primary strategic asset of the BiteFixes platform.

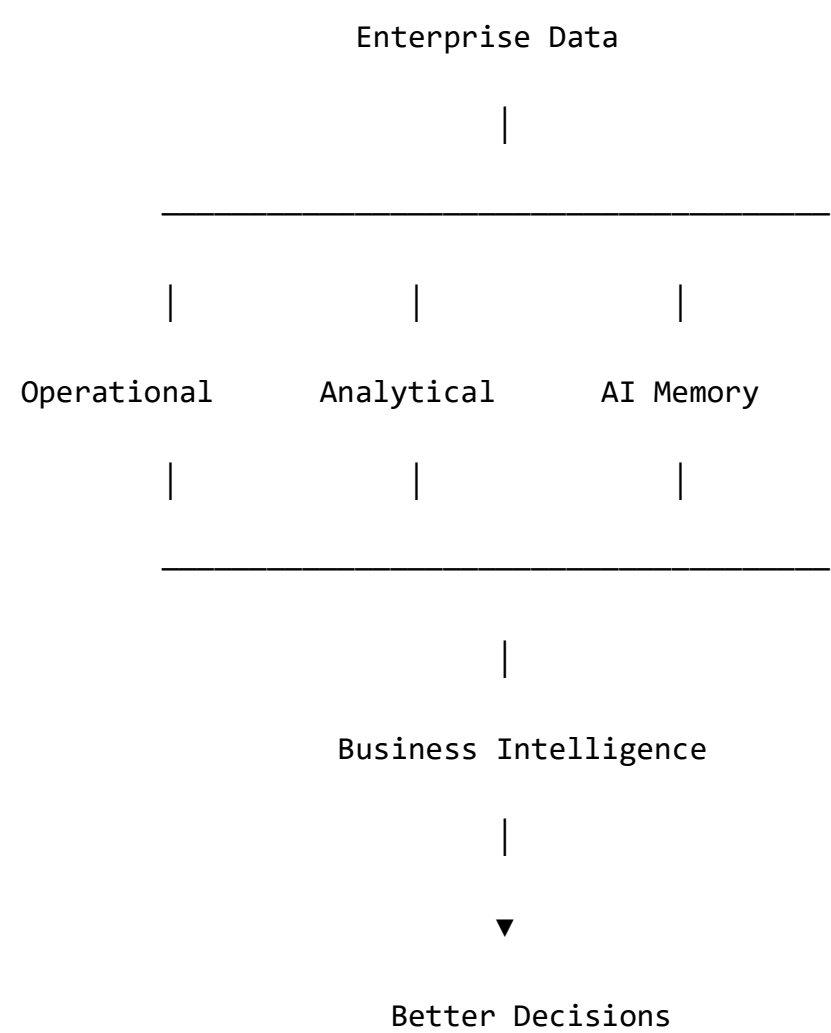
Every interaction generates information that contributes to:

- Operational efficiency

- Customer experience
- Artificial intelligence
- Predictive analytics
- Business insights
- Automation

The architecture treats data as a shared enterprise resource while maintaining strict tenant isolation.

Enterprise Data Vision



14.3 Enterprise Data Principles

Principle 1 — Single Source of Truth

Each business entity has a single authoritative source.

Examples:

Entity	Authoritative Source
Customer	customers
Device	devices
Ticket	tickets
Service	services
Company	companies

Duplicate business information should be avoided.

Principle 2 — Data Ownership

Every business entity has a defined owner.

Example

Company

↓

Owns Customers

↓

Owns Devices

↓

Owns Tickets

↓

Owns Knowledge



Owns AI Context

Ownership defines governance, permissions, and lifecycle.

Principle 3 — Data Integrity

Integrity is maintained through:

- Primary keys
- Foreign keys
- Constraints
- Transactions
- Referential integrity
- Validation rules

Business rules must never rely solely on application logic.

Principle 4 — Data Minimization

Only the information necessary to deliver the service is collected.

Benefits:

- Better privacy
- Lower storage costs
- Simpler governance
- Improved compliance

Principle 5 — AI-Ready Data

Every operational dataset should be designed to support:

- AI reasoning
- Semantic search
- Intent detection
- Context generation
- Recommendation systems
- Future machine learning models

14.4 Data Domains

The platform organizes information into independent business domains.

Company Domain

Stores enterprise information.

Entities include:

Company

CompanySettings

CompanyBrand

CompanySubscription

CompanyConfiguration

Customer Domain

Stores customer-related information.

Customer

CustomerProfile

CustomerContact

CustomerPreferences

CustomerChannel

Device Domain

Stores information about customer devices.

Device

DeviceCategory

DeviceBrand

DeviceModel

DeviceStatus

DiagnosticHistory

RepairHistory

Service Domain

Defines the service catalog.

Service

ServiceCategory

ServicePrice

ServicePackage

ServiceWorkflow

Ticket Domain

Supports operational workflows.

Ticket

TicketStatus

TicketAssignment

TicketHistory

TicketComment

TicketAttachment

AI Domain

Contains information generated by Bitey.

Intent

Conversation

KnowledgeArticle

Memory

Embedding

PromptTemplate

AIEvent

AIResponse

ConfidenceScore

Administration Domain

Supports internal platform management.

User

Role

Permission

AuditLog

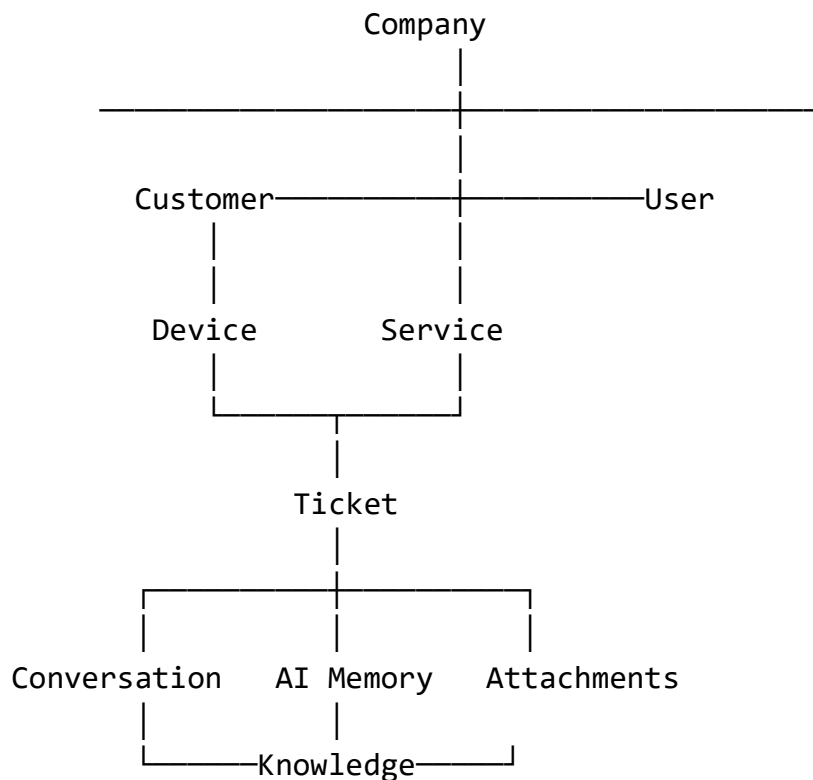
Notification

APIKey

Integration

14.5 Enterprise Data Model

The enterprise data model defines relationships among all business domains.



This model serves as the conceptual foundation for the physical PostgreSQL/Supabase schema.

14.6 Operational Data Store

The Operational Data Store (ODS) contains all data required for day-to-day platform operations.

Characteristics:

- Highly normalized
- ACID-compliant
- Transaction-oriented
- Low latency
- Real-time updates

Primary storage technology:

- PostgreSQL (Supabase)

Operational tables include:

companies

customers

devices

services

tickets

ticket_events

users

roles

permissions

conversations

knowledge_base

memory

notifications

All operational services read from and write to the ODS, ensuring transactional consistency across the platform.